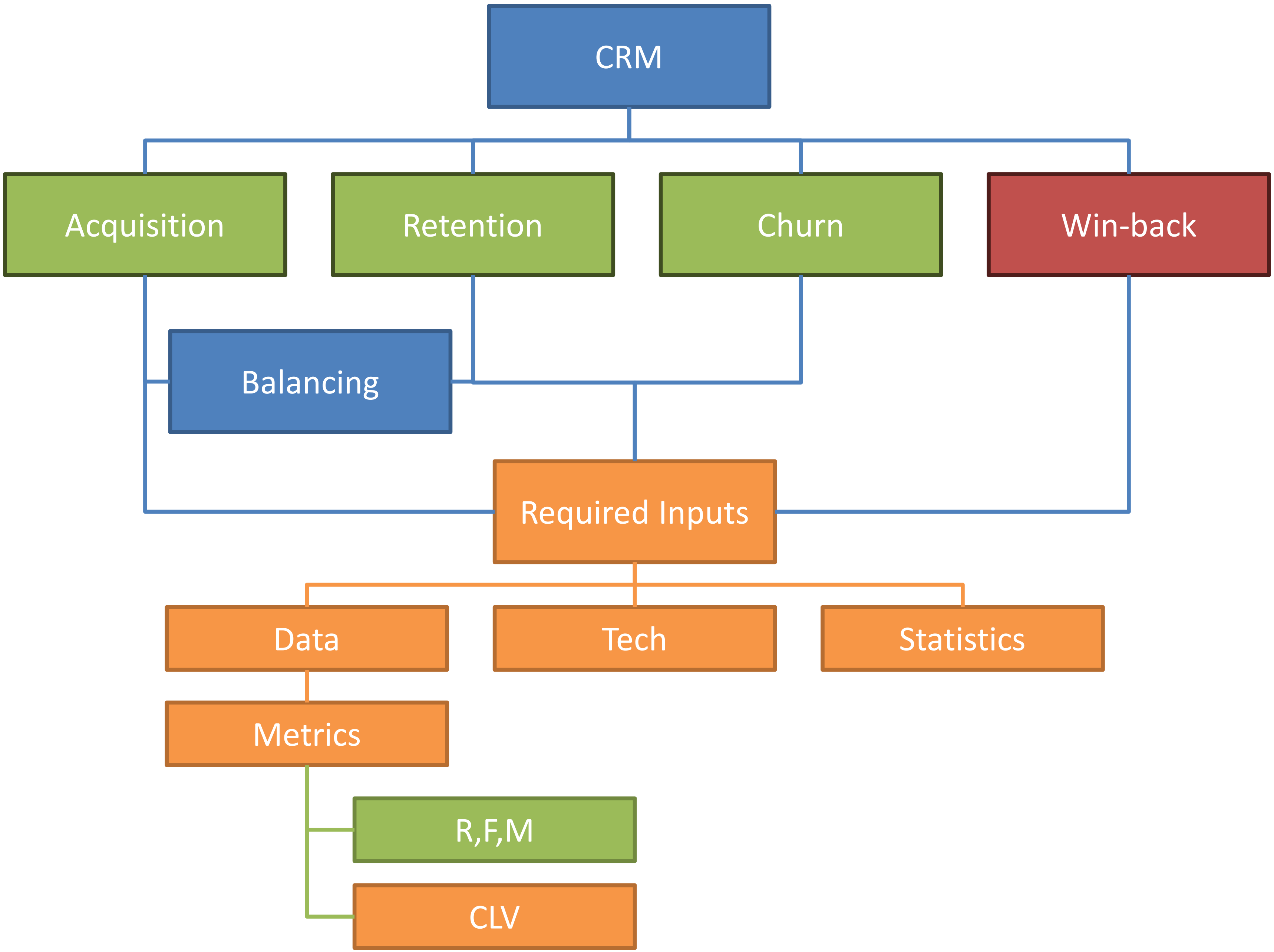
Contemporary Issues in Business Analysis
STA842/BWIB822
Customer Churn Modelling

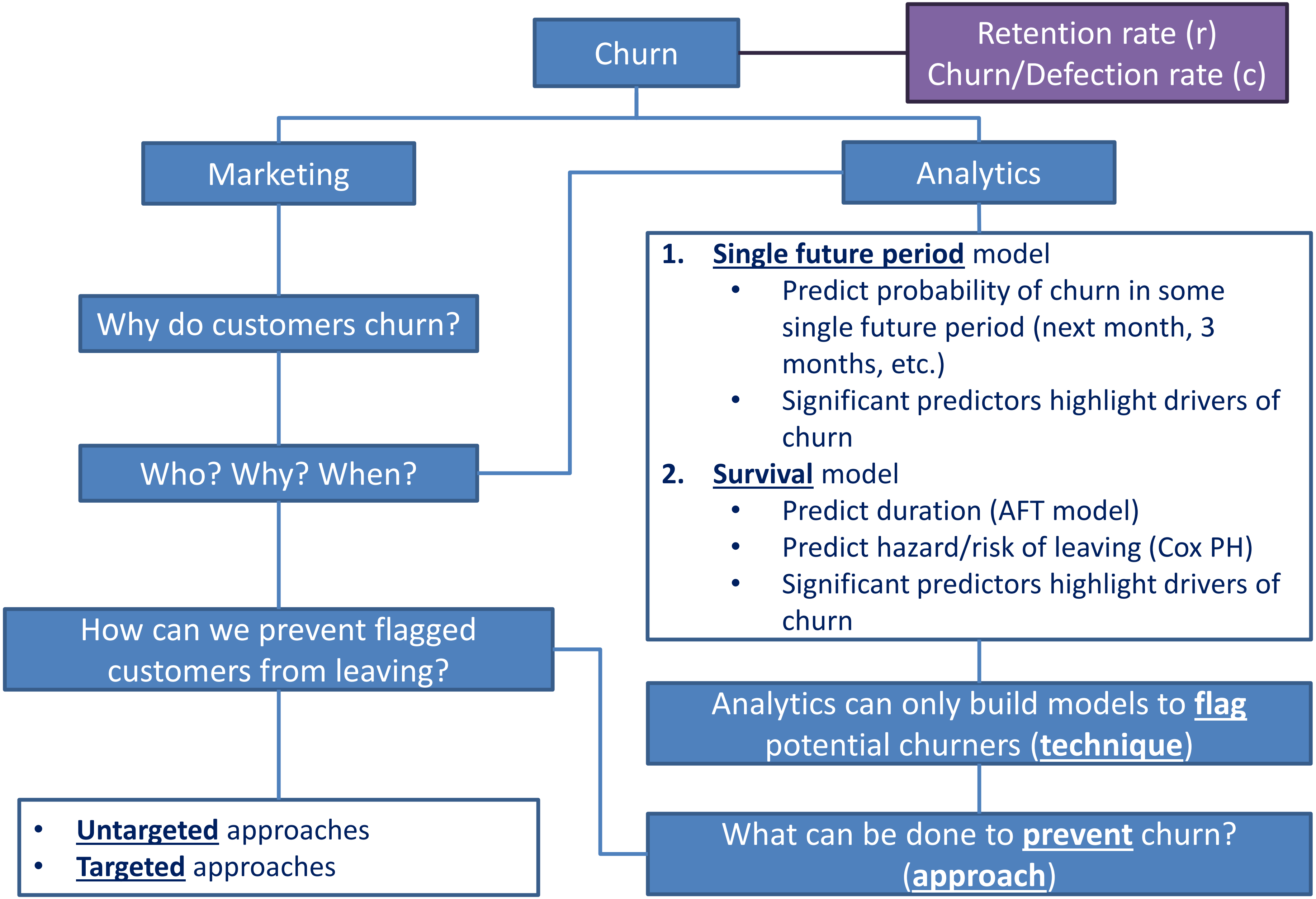
Chapter 24: Churn Management

Blattberg, R. C., Kim, B. D. & Neslin, S. A. (2008). *Database marketing*. New York: Springer Science+Business Media, LLC.

Chapter 6: Customer churn

Kumar, V. & Petersen, J. A. (2012). *Statistical Methods in Customer Relationship Management*. West Sussex: John Wiley & Sons, Ltd.


CRM
Acquisition
Retention
Churn
Win-back
Balancing
Required Inputs
Data
Tech
Statistics
Metrics
R,F,M
CLV


Churn
Retention rate (r)
Churn/Defection rate (c)
Marketing
Analytics
Why do customers churn?
Who? Why? When?
How can we prevent flagged customers from leaving?
Untargeted approaches
Targeted approaches
1. Single future period model
Predict probability of churn in some single future period (next month, 3 months, etc.)
Significant predictors highlight drivers of churn
2. Survival model
Predict duration (AFT model)
Predict hazard/risk of leaving (Cox PH)
Significant predictors highlight drivers of churn
Analytics can only build models to flag potential churners (technique)
What can be done to prevent churn? (approach)

Background

- **To date** we have considered **acquisition** and **retention** of most profitable customers
- Retention of customers is the goal, but **churn/attrition** always happens
  - Can be extremely **costly**!
- **Understanding and managing churn** are financially important!
- Today's seminar:
  1. **Why** discuss churn management?
  2. **Factors** that cause churn
  3. **Proactive** churn management
  4. **Incentives** to prevent churn

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WHY DISCUSS CHURN MANAGEMENT?

The Problem

- Using a **<u>simple retention model (Retention Part 3)</u>**, customer lifetime value (**<u>LTV</u>**) is:

$$LTV = \sum_{t=1}^{\infty} \frac{m_t \boxed{r}^{t-1}}{(1+\delta)^{t-1}}$$

  - m_t: customer's profit contribution in period t
  - r: retention rate   Focus of churn management
  - δ: discount rate
- **<u>Definition</u>** of churn:
  - **<u>Customer level</u>**: probability the customer leaves in a given period
  - **<u>Firm level</u>**: proportion of customer base that leaves in a given period

➢ Churn $= c = 1 - retention\ rate = 1 - r$

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The Problem

Two types of churn

1. **Involuntary** churn
   - **Company decides** to terminate relationship with customer
2. **Voluntary** churn
   - **Customer decides** to terminate relationship with company

     a) **Deliberate voluntary** churn
       - Customer dissatisfied
       - Customer received better offer

     b) **Incidental voluntary** churn
       - Customer no longer needs product
       - Customer moved to location not serviced by company

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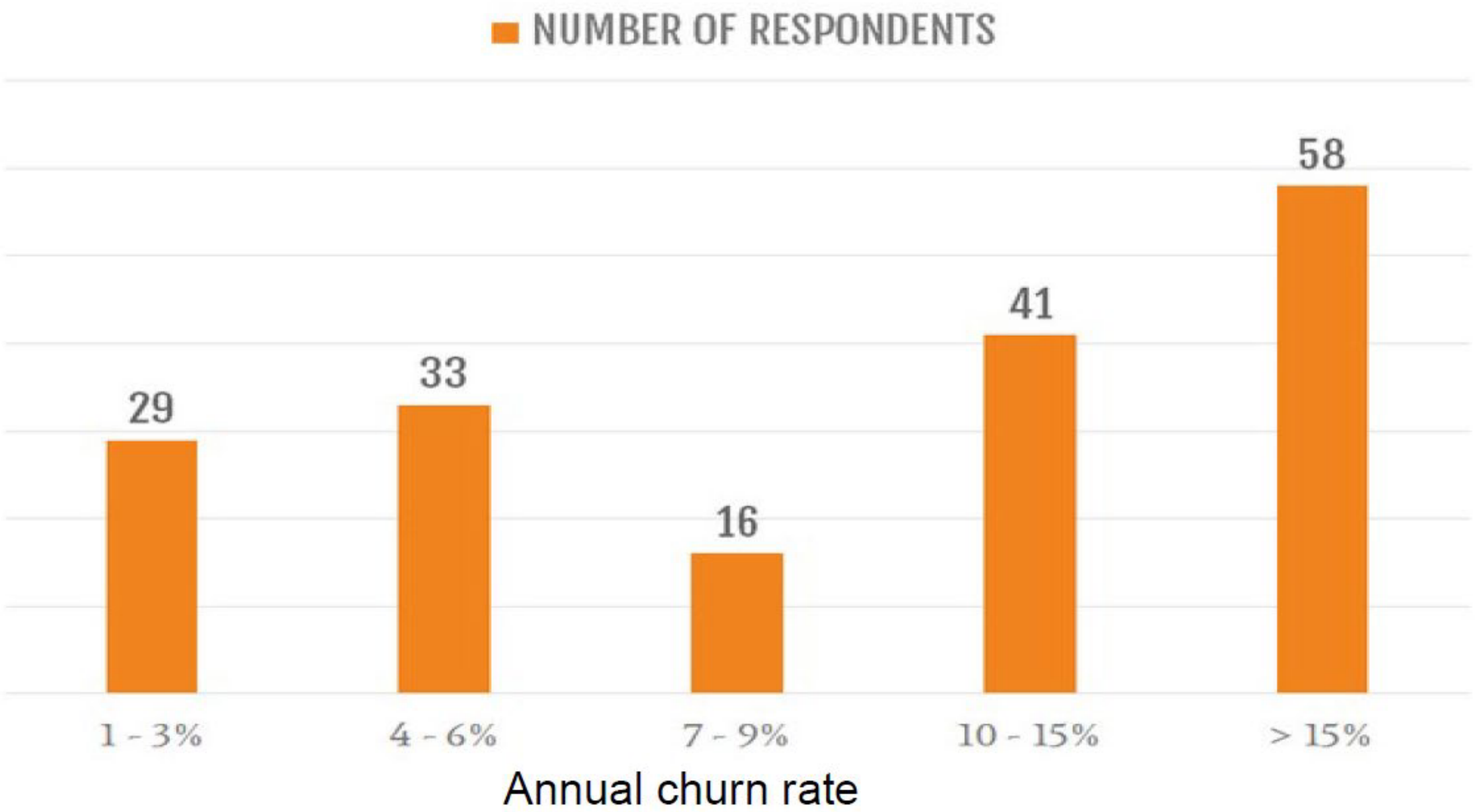
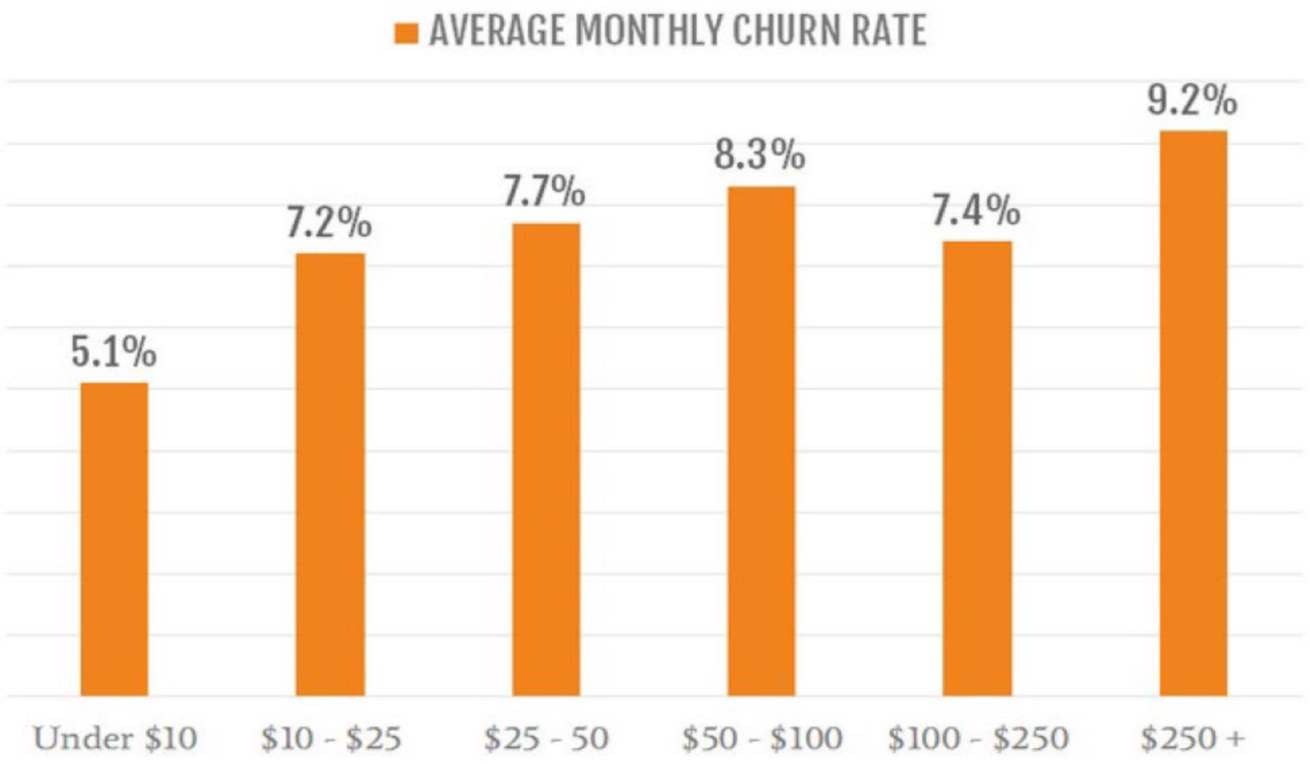
The Problem



Source: https://www.cobloom.com/blog/churn-rate-how-high-is-too-high

The Problem



$ amount reflects the average revenue per account
Source: https://www.cobloom.com/blog/churn-rate-how-high-is-too-high

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The Problem

- Table 24.1 (Blattberg et al., p. 609): **20% - 50%** of customers that start the year with a company, **leave** by year end
- **To illustrate** importance of such churn rates:

1. **Expected customer lifetime**
   - ➢ How long customer stays with company
   - ➢ $= \frac{1}{c}$, reciprocal of churn rate (see Appendix 24.1, Blattberg et al.)

Figure 24.1, p. 610, Blattberg et al.

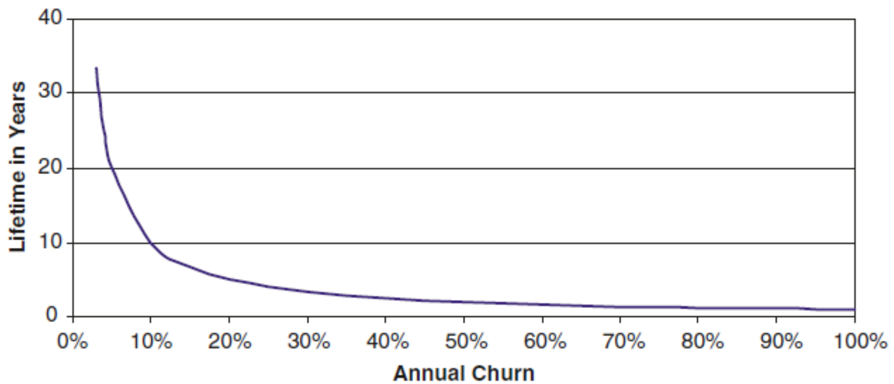


Fig. 24.1 Expected lifetime as function of churn rate (Equation 24.3).

Annual churn rate	50%	40%	30%	20%
Expected lifetime (years)	2	2.5	3.3	5

➢ Lower churn rate, longer lifetime

The Problem

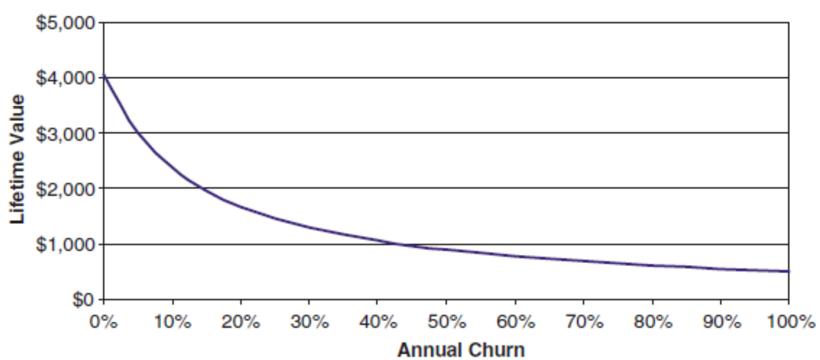
2. Customer lifetime value

- Lower churn rate, higher lifetime value
- Lower churn rate, higher ROI

$$LTV = \sum_{t=1}^{\infty} \frac{m_t \boxed{r}^{t-1}}{(1+\delta)^{t-1}} \Rightarrow LTV = \sum_{t=1}^{\infty} \frac{m\boxed{(1-c)}^{t-1}}{(1+\delta)^{t-1}} = \boxed{\frac{m(1+\delta)}{(\delta+c)}}$$

- m: annual profit contribution per customer
- c: annual churn rate
- δ: annual discount rate

Figure 24.2, p. 610, Blattberg et al. (Assuming $\delta = 14\%$ & $m = \$500$ per year)



Annual churn rate	50%	40%	30%	20%
Lifetime value ($ per customer)	891	1056	1295	1676
ROI (acq cost = $400)	123%	164%	224%	319%

FACTORS THAT CAUSE CHURN

Factors that Cause Churn

- The first step towards predicting and remedying churn is to **<u>understand the factors</u>** that cause or alleviate it

<table>
<tr><th>CUSTOMER SATISFACTION</th><th>SWITCHING COSTS</th><th>CUSTOMER CHARACTERISTICS</th></tr>
<tr><td>• Service quality
• Fit-to-needs
• Meeting expectations
• Price</td><td>• Physical
• Psychological</td><td>• Risk aversion
• Variety seeking
• Deal proneness
• Mavenism</td></tr>
<tr><th>MARKETING</th><th colspan="2">COMPETITION</th></tr>
<tr><td>• Reward programs
• Promotions
• Price
• Customized products</td><td colspan="2">• Within category
• Between category</td></tr>
</table>

See p. 611-615 in Blattberg et al. for discussion of other factors

Switching Costs

Low switching cost ⇒ Increased churn probability

<u>Two forms</u>:

1. Physical switching cost
   - Real inconvenience
   - E.g. “mobile number portability”
2. Psychological switching cost
   - Inertia
   - Brand pull
   - Familiarity

- **<u>Interesting example</u>**: number portability

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Customer Characteristics

Higher churn likelihood

- Risk takers
- Variety seekers
- Shopping mavens
- Deal prone customers
- Lower income customers

Lower churn likelihood

- Rural customers
- Higher income customers
- Married customers

Marketing Efforts

Effective marketing ⇒ Decrease churn

✓ Things found to **work**:

- Special services
- Loyalty programs
- Matching customers to the right price plans

PROACTIVE CHURN MANAGEMENT

Who’s likely to churn?

Why are they churning?

When are they likely to churn?

Proactive Churn Management

- **<u>Key</u>**: identifying customers likely to churn
  - ➢ **<u>Predictive modelling</u>**

- No data on causes of churn
- No psychographic data (risk-taking)
- Competitive activity

- Behavioural measures (R, F, M)
- Customer complaints
- Offers extended to the customer
- Price information
- Any retention efforts
- Etc.

Proactive Churn Management

Two types of churn models

- **Single future period**
  - Predictors available for one or more time periods before "churn period"
  - Churn period: where we observe whether customer has churned
- **Time series**
  - Observe predictors and churn simultaneously as they occur period to period
  - Collect potential predictors and whether churn has taken place in each period
  - Keep track of duration…

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Single Future Period Models

- **Example**: credit card industry
- Potential **predictors** (collected 6 months before target month)
  - Demographics (age, occupation, credit rating, etc.)
  - Product ownership (services offered by bank)
  - Product usage (R, F, M)

105 predictors

- **Target**: churn or not (observed in the target month)
- **Variable selection**: exploratory / stepwise
- **Model**: neural net
- **6 predictors** selected

Credit card balance month 6	Interest charged month 3	Purchase level month 6
Household age	Payment year-to-date month 2	Cash advances month 4

Single Future Period Models in Action: 1

The "Balance Transfer Flight Risk" Profile

- **<u>From predictors</u>:** High *Interest charged month 3* + High *Credit card balance month 6*.
- This customer is carrying a heavy balance and getting hit with high interest charges. They are highly susceptible to a competitor's "0% APR on Balance Transfers" promotional offer.
  - APR: annual percentage rate
  - Total annual cost of borrowing money, expressed as a percentage, including the interest rate, fees, and other charges

➢ **<u>Possible incentive</u>**: Proactively offer them a temporarily lowered APR for the next 6 months.

  ➢ Offer to convert their current high-interest balance into a fixed-fee, lower-interest installment plan. This retains the debt within your bank but lowers their monthly burden, removing the incentive to switch.

Single Future Period Models in Action: 1

"Balance Transfer Flight Risk" ⇒ Cost of False Negatives?

- **False negative**: the model predicts the customer will stay, so the bank does nothing, but the customer actually churns
- For a customer with a **high balance**, their expected future value **(CLV) is high** because they generate significant monthly interest revenue for the bank on this high balance.
- If the bank uses a default threshold like 0.5 to turn probabilities into churn predictions that would trigger interventions on their system, the model will probably generate too many false negatives. The bank would miss the chance to save a high CLV customer, taking a huge financial hit.
  - ➢ **Lower the threshold**, deliberately accepting more false positives (offering lower rates to customers who might have stayed anyway) just to ensure they do not commit a costly false negative.

Single Future Period Models in Action: 2

The "Lost Wallet Share" Profile

- **<u>From predictors</u>:** Decrease in *Purchase level month 6* + *Household age* demographic.
- The customer isn't necessarily closing the account right away, but they have stopped using your card for daily spending. They have likely moved a competitor's card to the "top of their wallet" because of better rewards tailored to their age/lifestyle.

➢ **<u>Possible incentive</u>**: Spend and Get Promotion.

  ➢ Offer a targeted, short-term incentive, such as "Earn 5,000 bonus points when you spend R2,000 on groceries this month." This forces them to pull your card back out of their wallet and re-establish the habit of using it.

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Single Future Period Models in Action: 2

“Lost Wallet Share" ⇒ Threshold adjustment?

- The **threshold** for action is directly proportional to how much the intervention costs.
- The proactive measure here – spend R2 000 and get 5 000 points – is very cost efficient for the bank. They will only incur a cost if the customer spends the money.
- Every time a customer swipes a credit card, the bank collects an **interchange fee** from the merchant (usually between 1.5% and 3% of the transaction). If the promotion requires the customer to spend R2,000 on groceries to get the bonus, the bank is actively generating R30 to R60 in new revenue. This interchange revenue offsets the cost of the reward points they are giving away.
- Because the cost is so low the bank can afford to give a promotion to a customer that wasn’t really going to churn. They can thus **accept more false positives**
  - ➢ **Lower the threshold**

Time Series Models

Chapter 6
Kumar & Petersen (2012)

Why and when is a customer likely to churn?

- **Example**: B2B firm, contractual basis
  - Churn observed
- **Modelling** focuses on whether or not customer defected in the **1st two years (730 days)** of the relationship
  - **Non-defects**: observe **censored lifetime** of 730 days
- **Aim**: understand difference between customers that have churned and customers that haven't left yet:
  - Determine **drivers of customer churn**
  - **Predict expected duration** of customers yet to churn
  - Determine **predictive accuracy** of model

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Time Series Models

List of variables:

Variable	
Customer	*Customer number (from 1 to 500)*
Duration	*The time in days that the acquired prospect has been or was a customer, right-censored at 730 d*
Censor	*1 if the customer was still a customer at the end of the observation window, 0 otherwise*
Avg_Ret_Exp	*Average number of dollars spent on marketing efforts to try and retain that customer per month*
Avg_Ret_Exp_SQ	*Square of the average number of dollars spent on marketing efforts to try and retain that customer per month*
Total_Crossbuy	*The total number of categories the customer has purchased from during the customer's lifetime*
Total_Freq	*The total number of purchase occasions the customer had with the firm in the customer's lifetime*
Total_Freq_SQ	*The square of the total number of purchase occasions the customer had with the firm in the customer's lifetime*
Industry	*1 if the prospect is in the B2B industry, 0 otherwise*
Revenue	*Annual sales revenue of the prospect's firm (in millions of dollars)*
Employees	*Number of employees in the prospect's firm*

- **Model** used: accelerated failure time (AFT) model

The AFT Model

- **Parametric** alternative to Cox Proportional Hazard
- Examines predictors' effects on event times in **censored data** regression
  - ➢ Looks at 1st two years of relationship (730 days)
- **Model**:

$$\ln(Duration_i) = X_i'\beta + \sigma\varepsilon_i$$

  - $\ln(Duration_i)$: natural log of customer i duration (**log-normal dist**)
    - ➢ **Other** possible distributions: Exponential, Weibull, log-logistic
  - X_i: predictors for customer i
  - β: coefficients
  - σ: scale parameter
  - ε_i: random error

The AFT in SAS

SAS code:

```
proc lifereg data=work.customer_churn;
    class industry;
    model duration*censor(1) = avg_ret_exp avg_ret_exp_sq
    industry revenue employees total_crossbuy total_freq
    total_freq_sq / dist=lognormal;
    output out = work.churn_pred p=pred_churn xbeta=xbeta_churn;
run; quit;
```

Model output:

Analysis of Maximum Likelihood Parameter Estimates								
Parameter		DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept		1	5.7417	0.0505	5.6427	5.8406	12927.8	<.0001
Avg_Ret_Exp		1	0.0089	0.0008	0.0072	0.0106	110.22	<.0001
Avg_Ret_Exp_SQ		1	-0.0002	0.0000	-0.0002	-0.0002	734.41	<.0001
Industry	0	1	0.0283	0.0191	-0.0090	0.0657	2.21	0.1372
Industry	1	0	0.0000	.	.	.	.	.
Revenue		1	0.0035	0.0006	0.0024	0.0046	38.53	<.0001
Employees		1	0.0004	0.0000	0.0004	0.0005	295.83	<.0001
Total_Crossbuy		1	0.0977	0.0066	0.0847	0.1107	217.41	<.0001
Total_Freq		1	0.0275	0.0069	0.0140	0.0409	16.02	<.0001
Total_Freq_SQ		1	-0.0009	0.0003	-0.0015	-0.0003	7.86	0.0050
Scale		1	0.1582	0.0074	0.1444	0.1733		

The AFT in SAS

- How do **changes in the drivers of customer duration** affect expected customer duration?
  - $\frac{T(X_i+\delta)}{T(X_i)} = exp(\delta\beta)$
  - $T(\cdot)$: duration model
  - X_i: value of independent variable for customer i
  - δ: change in X_i
    - $\delta = 1 \Rightarrow exp(\beta)$
- **Computed time ratios** for significant predictors:

Variable	Duration Ratio
Average retention exp	$\exp(0.0088 - 0.0004 \cdot AvgRetExp)$
Total crossbuy	1.103
Total frequency	$\exp(0.027 - 0.002 \cdot TotalFreq)$
Revenue	1.004
Employees	1.0004

The AFT in SAS

SAS code:

```
/* Predictive accuracy */

proc means data=work.churn_pred mean;
    var duration;
    where duration < 730;
run;

data work.churn_pred1;
    set work.churn_pred;
    ad = abs(duration - pred_churn);
    ad1 = abs(502.68 - duration);
    if pred_churn < 730 then pred_ch = 1; else pred_ch = 0;
    if duration < 730 then act_ch = 1; else act_ch = 0;
run; quit;
```

```
/* Duration prediction */

proc sql;
    select mean(duration) as mean_dur,
    mean(ad) as mad_dur,
    mean(ad1) as mean_dur_rand
    from work.churn_pred1 where censor=0;
quit;
/* Churn prediction */

proc freq data=work.churn_pred1;
    table pred_ch*act_ch;
run; quit;
```

Accuracy output:

Duration

mean_dur	mad_dur	mean_dur_rand
502.681	76.6954	142.3593

Churn

Table of pred_ch by act_ch

Frequency / Percent / Row Pct / Col Pct	act_ch 0	act_ch 1	Total
pred_ch 0	231 46.20 85.87 86.19	38 7.60 14.13 16.38	269 53.80
pred_ch 1	37 7.40 16.02 13.81	194 38.80 83.98 83.62	231 46.20
Total	268 53.60	232 46.40	500 100.00

Proactive Churn Management Last Words

- In summary (p. 622 in Blattberg et al.)

"Churn modelling requires an approach, not just a statistical technique, and that while different approaches can be successful, not all are successful"

INCENTIVES TO PREVENT CHURN

Managerial Approaches to Reducing Churn

Untargeted approaches

- Increase customer satisfaction / switching costs
- Improved products, advertising or loyalty programs

Targeted approaches

- Try to rescue customers identified as likely to churn

Managerial Approaches to Reducing Churn

Two types of **targeted approaches**

- **Reactive**
  - Wait for customers to identify themselves as likely to churn
  - E.g. calls to cancel service
- **Proactive**
  - Advanced identification of likely churners
  - Diagnose reason for potential churn
  - Takes appropriate action to prevent churn

Reactive Churn Management

Perfect/near-perfect prediction of churn

- Always addressing the correct customers with regards to churn management
- Company **can afford significant incentive** to prevent churn

Large incentive costs a lot in the short term

- Can be **costly due to the size** of the incentive needed to convince a leaving customer to stay
- May **incentivise** customer to call whenever customer receives competitive offer

Proactive Churn Management

Relies on a predictive model to flag possible churners

- Predictive accuracy depends on model quality
- Predictions often poor due to lack of data
- Could be missing out on profitable customers leaving the business without your knowledge

Might **stimulate satisfied customers to consider churning**

- Company **cannot spend too much** per incentive to limit potential waste since you **might make an offer to a customer that wasn't going to churn**

Framework for Proactive Churn Management

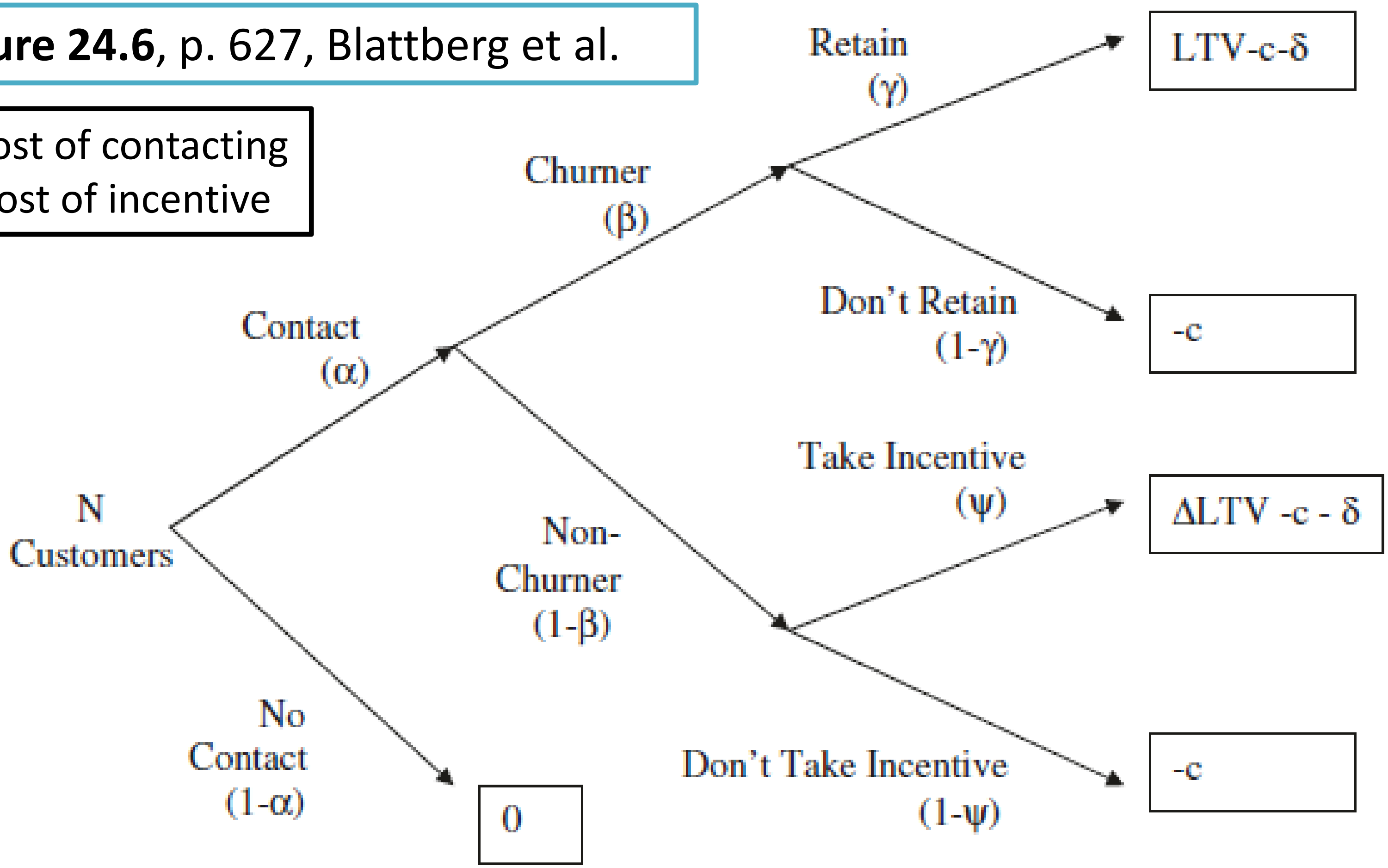
- Potential **trade-offs** to proactive churn management:
  - **Predictive accuracy** of model
  - **Effectiveness** of **offers** made
  - **Potential wastage** of offers made to non-churners

- These issues can be quantified by a **profitability framework**
- Profitability framework can be used in two ways:
  1. Will a **proposed offer** result in some **increase** in **revenue**?
  2. **How much should be offered** if the business seeks **x% increase** in **ROI**?

Framework for Proactive Churn Management

Figure 24.6, p. 627, Blattberg et al.

c: cost of contacting
δ: cost of incentive



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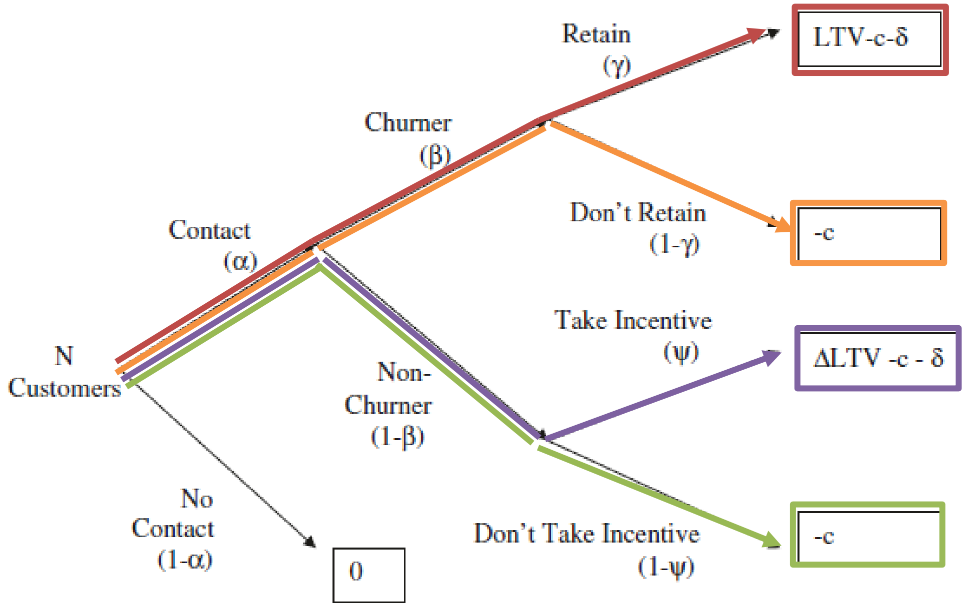
Framework for Proactive Churn Management

Variable definitions

- N: total nr of customers
- α: $P(contact)$ in churn management program
- β: $P(churn|contact)$
- γ: $P(rescue|churn)$
- ψ: $P(incentive|non-churner)$
- Δ: % increase in LTV of non-churners that take incentive
- c: cost of contact
- δ: cost of incentive
- LTV: lifetime value of customer

Framework for Proactive Churn Management

Figure 24.6, p. 627, Blattberg et al.



➢ **Total profit** of framework:

$$\begin{aligned}\Pi &= N\{\boxed{\alpha\beta\gamma(LVC - c - \delta)} + \boxed{\alpha\beta(1-\gamma)(-c)} + \boxed{\alpha(1-\beta)\psi(\Delta LVC - c - \delta)} \\ &\quad + \boxed{\alpha(1-\beta)(1-\psi)(-c)}\} \\ &= N\alpha\{(\beta\gamma + (1-\beta)\psi\Delta)LVC - \delta(\beta\gamma + (1-\beta)\psi) - c\} \qquad (24.8)\end{aligned}$$

Optimal Incentive Value?

- From **Equation 24.8**: **Total profit** of churn management program

$$\begin{aligned}\Pi &= N\{\alpha\beta\gamma(LVC - c - \delta) + \alpha\beta(1-\gamma)(-c) + \alpha(1-\beta)\psi(\Delta LVC - c - \delta) \\ &\quad + \alpha(1-\beta)(1-\psi)(-c)\} \\ &= N\alpha\{(\beta\gamma + (1-\beta)\psi\Delta)LVC - \delta(\beta\gamma + (1-\beta)\psi) - c\} \qquad (24.8)\end{aligned}$$

➢ **Total investment** in churn program (3rd and 4th terms):

$$N\alpha[\delta(\beta\gamma + (1-\beta)\psi) - c]$$

➢ **ROI**: total profit divided by total investment

$$\frac{N\alpha[(\beta\gamma + (1-\beta)\psi\Delta)LTV - \delta(\beta\gamma + (1-\beta)\psi) - c]}{N\alpha[\delta(\beta\gamma + (1-\beta)\psi) - c]}$$

➢ Set ROI requirement, $ROI > \rho$, and **solve for δ** (incentive cost)

  ➢ **Equation 24.9**, p. 628, Blattberg et al.

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Summary

Why discuss churn management?

- Customer base can become depleted (c > a)
- Churn rate can be higher than expected
- Most profitable customers tend to be most sensitive

Factors that cause churn

- (Lack of) switching cost
- Customer characteristics
- (Lack of) effective marketing efforts

Proactive churn management

- Single future period model (who, why)
- Survival model (who, why, when)

Incentives to prevent churn

- Targeted vs untargeted approaches
- Churn management framework

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Homework

Classwork (in groups)

Scheduled due date: 12 May

Discuss

Class test 2

Date and Time: 12 May at 11h (75 mins)

Scope on eFundi

Prep for 12 May

Blattberg et al. Chapter 5